

Canalicular ablation: where do the straight segments come from?
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PIXEL SIZE 0.13 um/px. PIXEL SIZE 0.13 UM/PIX, ROUNDED, UNCONFIRMED.
PRE-VALIDATION. A diagnosis, not a new method. No variant is called better.

Variants:
A current method, as is
B current method, gap bridging off
C ridge method, as is
D ridge method, gap bridging off
E ridge method, low cut 0.9 of the high cut
F ridge method, bridging off, single scale 3.0 px
G ridge method, bridging off, finer scales

Mean over the 8 images, and the change from variant A

measure	A	B	C	D	E	F	G
unsupported_fraction	0.06798	0.06699	0.121	0.1198	0.0904	0.09041	0.08882
bridged_fraction	0.0006067	0	0.0007291	0	0.005179	0	0
straight_run_fraction	0.221	0.2203	0.2635	0.2634	0.2577	0.25	0.2447
rectangle_count	0	0	0.25	0.25	0.125	0	0
connected_to_lacuna_fraction	0.336	0.3289	0.7191	0.7033	0.634	0.5489	0.5109
junction_count	529.8	526.8	876.1	871.6	727	728.9	723
thread_end_fraction	0.7641	0.7678	0.575	0.5817	0.623	0.6491	0.6747
width_median_px	6.041	6.041	7.408	7.408	6.768	6	5.743
field_length_density_per_px	0.0391	0.03904	0.04071	0.04064	0.0383	0.0398	0.04106
roots_per_cell	7.545	7.577	7.843	7.845	7.404	7.308	7.778
ring_length_r30_px_per_cell	321.5	321.2	339.2	338.4	316.1	320.4	330.6
skeleton_px	3.992e+04	3.985e+04	4.154e+04	4.148e+04	3.909e+04	4.062e+04	4.191e+04
n_bridges	18.12	0	16.25	0	70.5	0	0

Change from A, per cent

measure	A	B	C	D	E	F	G
unsupported_fraction	+0.0	-1.4	+78.0	+76.3	+33.0	+33.0	+30.7
bridged_fraction	+0.0	-100.0	+20.2	-100.0	+753.5	-100.0	-100.0
straight_run_fraction	+0.0	-0.3	+19.3	+19.2	+16.6	+13.1	+10.7
rectangle_count							
connected_to_lacuna_fraction	+0.0	-2.1	+114.0	+109.3	+88.7	+63.3	+52.0
junction_count	+0.0	-0.6	+65.4	+64.5	+37.2	+37.6	+36.5
thread_end_fraction	+0.0	+0.5	-24.8	-23.9	-18.5	-15.1	-11.7
width_median_px	+0.0	+0.0	+22.6	+22.6	+12.0	-0.7	-4.9
field_length_density_per_px	+0.0	-0.2	+4.1	+3.9	-2.0	+1.8	+5.0
roots_per_cell	+0.0	+0.4	+3.9	+4.0	-1.9	-3.1	+3.1
ring_length_r30_px_per_cell	+0.0	-0.1	+5.5	+5.3	-1.7	-0.4	+2.8
skeleton_px	+0.0	-0.2	+4.1	+3.9	-2.1	+1.8	+5.0
n_bridges	+0.0	-100.0	-10.3	-100.0	+289.0	-100.0	-100.0

Every image and variant

image	v	skeleton_px	unsupported_fraction	bridged_fraction	straight_run_fraction	rectangle_count	connected_to_lacuna_fraction	junction_count	width_median_px	field_length_density_per_px	roots_per_cell	n_bridges
542_z06	A	3.496e+04	0.05297	0.000715	0.2298	0	0.3743	375	6.325	0.03465	8.615	17
542_z06	B	3.49e+04	0.05167	0	0.2277	0	0.3861	374	6.325	0.03458	8.692	0
542_z06	C	3.899e+04	0.1338	0.000846	0.2611	0	0.6938	782	8	0.03864	11	18
542_z06	D	3.892e+04	0.1327	0	0.261	0	0.7043	770	8	0.03857	10.92	0
542_z06	E	3.674e+04	0.1027	0.004573	0.2569	0	0.6353	643	7.211	0.03641	9.923	64
542_z06	F	3.692e+04	0.08728	0	0.2408	0	0.5066	555	6	0.03659	8.385	0
542_z06	G	3.729e+04	0.08019	0	0.2342	0	0.4742	514	6	0.03695	9.308	0
542_z18	A	3.432e+04	0.05416	0.000612	0.2141	0	0.2993	395	6	0.03407	8.273	14
542_z18	B	3.427e+04	0.05305	0	0.2139	0	0.2843	395	6	0.03402	8.364	0
542_z18	C	3.765e+04	0.1234	0.000584	0.2689	0	0.6575	777	8	0.03737	8.546	12
542_z18	D	3.76e+04	0.1224	0	0.2688	0	0.6463	775	8	0.03732	8.546	0
542_z18	E	3.557e+04	0.0952	0.004611	0.2622	0	0.5639	658	7.211	0.03531	8.091	66
542_z18	F	3.669e+04	0.08988	0	0.2502	0	0.5074	645	6	0.03642	8.364	0
542_z18	G	3.698e+04	0.08305	0	0.2463	0	0.3389	573	6	0.03671	8.727	0
543-2	A	4.578e+04	0.07557	0.00059	0.206	0	0.4079	681	6	0.0447	7.583	21
543-2	B	4.572e+04	0.07483	0	0.2051	0	0.391	671	6	0.04464	7.583	0

543-2	C	4.507e+04	0.1045	0.000444	0.2601	0	0.8344	998	7.211	0.044	6.917	10
543-2	D	4.503e+04	0.1037	0	0.2597	0	0.8285	995	7.211	0.04396	6.917	0
543-2	E	4.278e+04	0.07763	0.004348	0.2525	0	0.7932	844	7.211	0.04177	6.583	65
543-2	F	4.434e+04	0.07668	0	0.2355	0	0.6623	846	6	0.04329	7.083	0
543-2	G	4.6e+04	0.07423	0	0.2279	0	0.6228	831	5.657	0.04492	7.417	0
543_3	A	4.642e+04	0.07652	0.000431	0.1997	0	0.2573	698	6	0.04514	6.889	14
543_3	B	4.637e+04	0.07591	0	0.1994	0	0.2467	695	6	0.04509	6.889	0
543_3	C	4.616e+04	0.1129	0.00052	0.2571	1	0.7952	1031	7.211	0.04489	7.111	13
543_3	D	4.61e+04	0.112	0	0.2571	1	0.7094	1030	7.211	0.04483	7.111	0
543_3	E	4.346e+04	0.0836	0.002761	0.2469	0	0.6558	849	7.211	0.04226	6.556	48
543_3	F	4.538e+04	0.08427	0	0.2447	0	0.503	886	6	0.04413	6.556	0
543_3	G	4.693e+04	0.07805	0	0.2341	0	0.3974	893	5.657	0.04564	6.889	0
543_z13	A	4.503e+04	0.07166	0.000444	0.2048	0	0.3864	643	6	0.0438	6.727	19
543_z13	B	4.498e+04	0.07098	0	0.2041	0	0.3792	641	6	0.04376	6.818	0
543_z13	C	4.411e+04	0.09861	0.000839	0.2537	0	0.6851	904	7.211	0.04291	6.364	20
543_z13	D	4.404e+04	0.09731	0	0.2534	0	0.674	898	7.211	0.04283	6.455	0
543_z13	E	4.157e+04	0.0721	0.006038	0.2509	0	0.6518	746	6.325	0.04043	5.636	84
543_z13	F	4.35e+04	0.07346	0	0.2424	0	0.5281	752	6	0.04232	6.455	0
543_z13	G	4.594e+04	0.07715	0	0.2435	0	0.519	839	5.657	0.04468	7	0
682_z08	A	3.527e+04	0.07412	0.000851	0.2446	0	0.4099	480	6	0.03465	10	24
682_z08	B	3.518e+04	0.07267	0	0.2432	0	0.4106	478	6	0.03457	10	0
682_z08	C	3.749e+04	0.1339	0.001147	0.2724	1	0.696	784	7.211	0.03683	10.62	22
682_z08	D	3.741e+04	0.1324	0	0.2727	1	0.6888	779	7.211	0.03675	10.62	0
682_z08	E	3.476e+04	0.09523	0.007509	0.2702	1	0.6428	645	6.325	0.03415	10.62	84
682_z08	F	3.678e+04	0.1054	0	0.2609	0	0.6153	678	6	0.03613	9.625	0
682_z08	G	3.84e+04	0.113	0	0.2657	0	0.6096	687	5.657	0.03773	10.25	0
682_z23	A	3.917e+04	0.06794	0.000664	0.2285	0	0.3067	481	6	0.0383	6.727	19
682_z23	B	3.911e+04	0.06684	0	0.2285	0	0.301	480	6	0.03824	6.727	0
682_z23	C	4.24e+04	0.1361	0.000613	0.2614	0	0.6997	913	7.211	0.04146	6.818	14
682_z23	D	4.234e+04	0.135	0	0.2616	0	0.6908	910	7.211	0.0414	6.818	0
682_z23	E	3.98e+04	0.1038	0.005628	0.2554	0	0.6633	748	6.325	0.03892	6.636	72
682_z23	F	4.157e+04	0.1074	0	0.2555	0	0.6215	769	6	0.04065	6.727	0
682_z23	G	4.282e+04	0.1074	0	0.2487	0	0.6366	763	5.657	0.04187	7	0
682_z29	A	3.837e+04	0.07087	0.000547	0.2403	0	0.2464	485	6	0.03748	5.545	17
682_z29	B	3.83e+04	0.06998	0	0.2403	0	0.2327	480	6	0.03741	5.545	0
682_z29	C	4.048e+04	0.1248	0.00084	0.2735	0	0.691	820	7.211	0.03955	5.364	21
682_z29	D	4.039e+04	0.1231	0	0.2732	0	0.6845	816	7.211	0.03946	5.364	0
682_z29	E	3.807e+04	0.09292	0.005963	0.2666	0	0.4661	683	6.325	0.03719	5.182	81
682_z29	F	3.977e+04	0.09886	0	0.27	0	0.447	700	6	0.03886	5.273	0
682_z29	G	4.094e+04	0.0975	0	0.2571	0	0.4884	684	5.657	0.04	5.636	0

pixel size 0.13 um/px, rounded, unconfirmed.

Pre-validation: no number here has been checked against a manual count.

A diagnosis, not a new method. No variant is better than another; this says what changed.

unsupported_fraction: share of skeleton pixels NOT above the image's own lower multi-Otsu cut of the flattened intensity, the same cut the current method uses as its high threshold.

straight_run_fraction: share of skeleton pixels inside a perfectly horizontal or vertical run of at least 8 px (about 1.04 um).

rectangle_count: independent four-node loops of the cleaned graph whose four sides are all axis-aligned to within 1 px (networkx.cycle_basis).

The verification images draw OWNED skeleton only, so threads no lacuna owns are not shown.